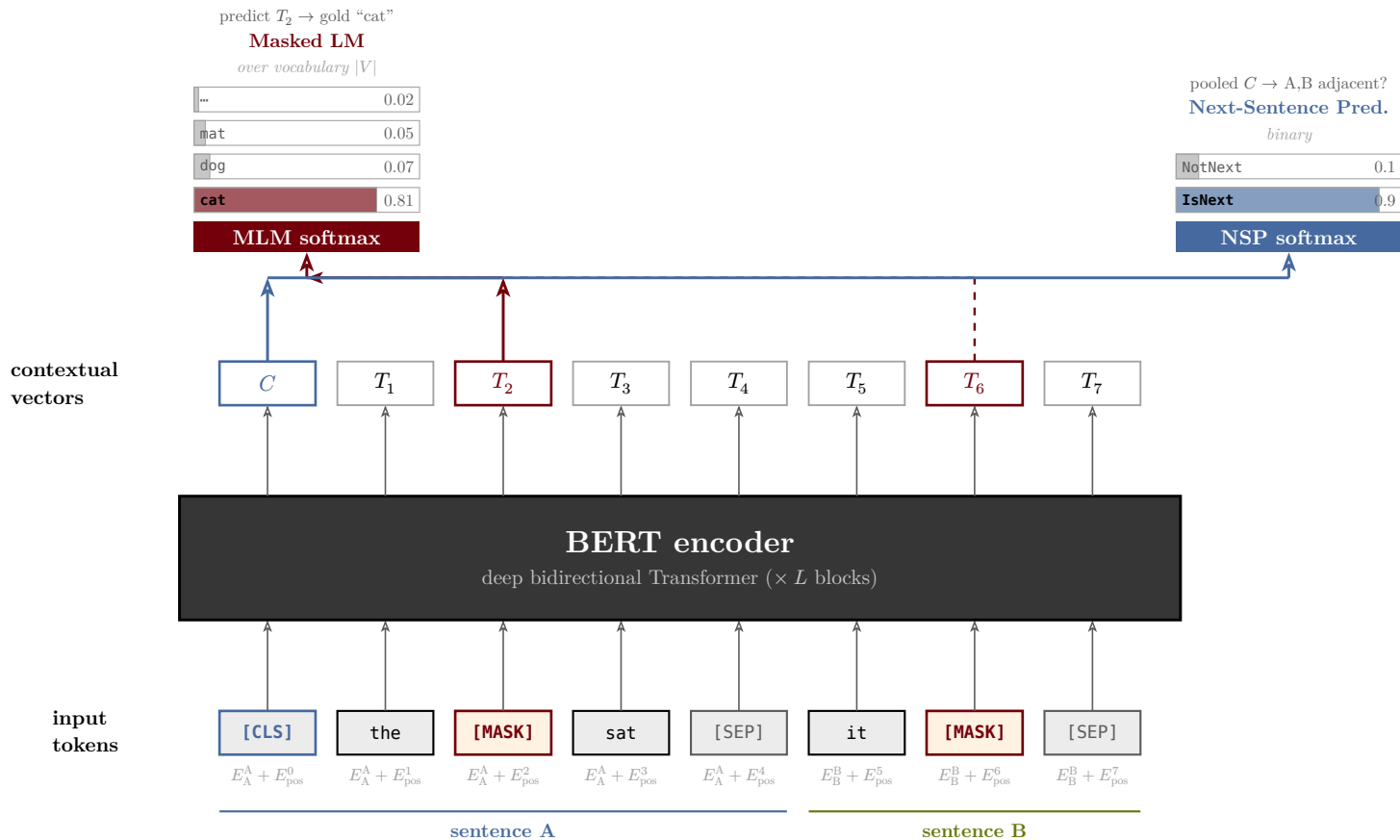


BERT pretraining: Masked-LM + Next-Sentence Prediction

corrupt $\sim 15\%$ of tokens with **[MASK]** · one bidirectional encoder · two read-out heads



Only the masked positions (T_2, T_6) are scored by the MLM softmax; only the $[CLS]$ vector C feeds the NSP softmax. Loss = MLM cross-entropy + NSP cross-entropy.